

Executive Summary & Conclusion

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This report summarises my capstone project, where I implemented and compared GLLiM, BLLiM, and GLLiM-Lw models for nonlinear regression using high-dimensional data. I started by generating synthetic datasets from three nonlinear functions (f , g , h) under four covariance structures: Independent, Block, Factor, and Toeplitz. The models were evaluated using RMSE across 100 runs. I also included Random Forest, SVM, MARS, and SIR models for comparison. I then reproduced Figure 2 and Table A.1 from Blein-Nicolas et al. (2024). In Week 8, I applied the models on real maize proteomic data to predict two drought-related traits (LAI.WD and WU.WD), evaluated using 10-fold cross-validation. I replicated Table 1 and generated Figures 3 and 4 to show top protein features from BLLiM. In Week 10, I implemented BLLiM and GLLiM-Lw fully in Python. Results showed BLLiM performed best with block covariance, GLLiM worked well with independent structure, and GLLiM-Lw provided a good trade-off between performance and interpretability. This project highlights that model choice should consider data structure. Limitations include the computational cost of EM-based models and the complexity of R-based workflows. Future work should focus on Python-based scalable implementations and feature reduction methods for better usability (Blein-Nicolas et al., 2024).

Conclusion

In conclusion, this project successfully demonstrated the use of GLLiM, BLLiM, and GLLiM-Lw for high-dimensional nonlinear regression tasks on both synthetic and real biological data. The evaluation showed that BLLiM is highly effective when data contains latent block structures, while GLLiM is better for simpler, independent cases. GLLiM-Lw provided a balance of interpretability and accuracy. However, computational efficiency remains a limitation, particularly with large-scale data. It is recommended that future developments focus on Python-native implementations, improved runtime efficiency, and incorporation of automatic feature selection to support real-time predictive applications.

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References

Blein-Nicolas, M., Nguyen, T. T., Gelly, M., & Besse, P. (2024). Nonlinear network-based quantitative trait prediction from biological data. **Journal of the Royal Statistical Society: Series C (Applied Statistics)**. <https://doi.org/10.1111/rssc.12668>